

Pressure Transient Analysis for Wells With Horizontal Drainholes

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ABSTRACT

Drainholes have been drilled in several areas of the world and there is a need to understand these drainage systems for accurate well test analysis. This study presents an analytical solution for the transient pressure response of a uniform flux horizontal drainhole in an anisotropic reservoir of finite thickness. The solution also applies for a reservoir with multiple drainholes in a vertical array. The solution shows that there are two possible types of transient pressure behavior depending on the length of the drainhole relative to the height of the reservoir. If the drainhole is short, initial radial flow perpendicular to the drainhole axis is followed by a transition to a pseudo-radial flow period. If the drainhole length is long relative to the reservoir height, the initial radial flow period ends rapidly and transient pressure behavior becomes identical to that of a uniform flux vertical fracture.

The pressure transient response for multiple drainholes is identical to the single drainhole solution if dimensionless variables are defined relative to the number of drainholes. Consequently, the pressure response of a uniform flux vertical fracture can also be approximated by a vertical array of drainholes. The pressure response for infinite conductivity drainholes is also suggested by analogy to the infinite conductivity vertical fracture solution.

Log-log type curves are presented for various drainhole radii and can be used in the conventional manner to determine reservoir characteristics including directional permeability or drainhole half length. Short and long time approximations are presented along with appropriate time limits. Finally, conditions for greater productivity than with vertical wells or hydraulic fractures are presented.

INTRODUCTION

Drainholes are horizontal bores drilled from a vertical wellbore to increase production. The purpose of this study is to present a means for understanding these systems using pressure transient analysis. Recently, many applications of horizontal drainholes have been attempted or suggested.¹⁻²⁴ This increased popularity is due mainly to attractive economics and better drilling technology.¹ Applications for drainholes includes: productivity index increases, injectivity index increases, greater sweep efficiencies by line flooding from one drainhole to another, water and gas coning reduction⁵⁻⁸, penetration of narrowly missed target zones, intersection of naturally occurring vertical fractures and reduced wellbore damage.¹

The available drainhole literature can be categorized by areas of operational feasibility, productivity increases, and EOR applications. The operational feasibility literature documents the development of flexible drill collars with better directional drilling.¹⁻⁴ These

References and illustrations at end of paper.

technological advances have reduced the amount of risk involved in drilling horizontal drainholes. Areas of the world where horizontal bores have been drilled with positive results include California,²⁵ Texas,⁵⁻⁸ France,^{4,9} Italy²⁶ and the USSR.¹⁰⁻¹²

The first studies on drainhole productivity using potentiometric models were presented in the mid-1950's by Perrine,²⁷ Roemershauser and Hawkins,²⁸ and Landrum and Crawford.²⁹ These three studies concluded that (1) drainhole length is of primary importance to productivity, (2) additional drainhole footage is more effective on a single well than are additional drainholes, and (3) increases in productivity are proportional to the combined length of all drainholes in the system until a point is reached where additional benefit per added foot begins to diminish. The conclusions of these reports are explained by the drainhole pseudo skin factor presented in this study.

Publications have been presented in the early 1980's that give analytical expressions and field results for determining expected productivity increases.^{9,30-32} The studies showed that horizontal bores were most advantageous for:

- tight reservoirs, specially if vertical fractures are suspected
- thin beds
- thin oil columns or a small density difference between water and oil
- soft formations such as chalk that are likely to collapse

The third area of drainhole literature is EOR applications of drainholes. The primary work in this area is from Canadian production of heavy crudes.¹⁴⁻¹⁷ Suggested or applied uses include continuous steam injection, cyclic injection production techniques, production of tight chalk formations and gravity drainage of steam-heated oil through parallel drainholes.¹⁴⁻²⁴

Although no published studies on the pressure transient analysis of horizontal drainholes currently exist, several studies in the pressure transient analysis literature are applicable to this study. First is the partially-penetrating uniform flux fracture solution presented analytically by Raghavan et al.³³ The general vertical fracture solution presented analytically for infinite conductivity and uniform flux fractures by Gringarten et al.³⁴ will be used to find similarities with the horizontal drainhole pressure transient response. A third study of interest is the pressure transient response for a slanted wellbore presented by Cinco.³⁵ Limiting cases of these three studies approximate the drainhole case and can be used for checking the drainhole solution.

PROBLEM DESCRIPTION

A solution to the pressure transient response of horizontal drainholes was derived using instantaneous source functions together

with the Newman product method. The combined application of these two methods for petroleum engineering problems was presented by Gringarten and Ramey³⁶.

A physical description of the problem is illustrated in Fig. 1. A drainhole is represented as a constant-rate line source of length $2x_f$ in a reservoir of height h with impermeable upper and lower boundaries. The reservoir is infinite in the x and y directions, with directional permeabilities of k_x , k_y , and k_z . The drainhole is located at height z_w above the bottom of the reservoir. The center of the drainhole is defined by x_w and y_w , the distances from the origin in the x and y directions respectively.

The equation governing flow is the diffusivity equation as derived from the continuity equation, an equation of state, and Darcy's Law^{37,38}:

$$\eta_x \frac{\partial^2 p(M,t)}{\partial x^2} + \eta_y \frac{\partial^2 p(M,t)}{\partial y^2} + \eta_z \frac{\partial^2 p(M,t)}{\partial z^2} - \frac{\partial p(M,t)}{\partial t} = 0 \quad (1)$$

where:

$$\eta_j = \frac{k_j}{\phi \mu c}, \quad j=x,y,z \text{ or } r \quad (2)$$

The instantaneous source function for the drainhole configuration shown in Fig. 1 can be obtained using the Newman product method³⁹. If a source-reservoir system can be represented as the intersection of three one-dimensional source-reservoir systems, then the instantaneous uniform flux source function is defined by the product of the three one-dimensional instantaneous source functions.

Many instantaneous source functions useful in pressure transient analysis have been presented previously by Gringarten and Ramey³⁶. These instantaneous source functions were used to generate the instantaneous source function for the drainhole-reservoir configuration. The system shown in Fig. 1 can be thought of as the intersection of three one-dimensional systems as shown in Fig. 2.

The following dimensionless variables were used.

$$x_D = \left(1 + \frac{x - x_w}{x_f}\right) \sqrt{\frac{k_y}{k_x}} - 1 \quad (3)$$

$$y_D = \frac{y - y_w}{x_f} \quad (4)$$

$$z_D = \frac{z - z_w}{x_f} \sqrt{\frac{k_y}{k_z}} \quad (5)$$

$$z_{wD} = \frac{z_w}{h} \quad (6)$$

$$L_D = \frac{x_f}{h} \sqrt{\frac{k_z}{k_y}} \quad (7)$$

$$t_D = \frac{k_y t}{\mu \phi c x_f^2} \quad (8)$$

Appendix A presents a derivation of the analytical solution given by Eq. (9).

$$p_D(x_D, y_D, z_D, L_D, t_D) = \frac{\sqrt{\pi}}{4} \int_0^{t_D} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-n^2 \pi^2 L_D^2 t_D' \right] \cos n \pi z_{wD} \right. \\ \left. \cdot \cos n \pi (z_D L_D + z_{wD}) \right] \left[\operatorname{erf} \frac{(1+x_D)}{2\sqrt{t_D'}} + \operatorname{erf} \frac{(1-x_D)}{2\sqrt{t_D'}} \right] \left[\frac{\exp \left[-\frac{y_D^2}{4t_D'} \right]}{\sqrt{t_D'}} \right] dt_D' \quad (9)$$

The analytical solution can not be expressed by known functions except for some limiting cases. To obtain solutions for a desired location or system configuration, it was necessary to evaluate the solution using Simpson's Rule. A computer program was obtained from Roland N. Horne which was used previously by Horne and Temeng⁴⁰. The program uses Simpson's Rule to evaluate definite integrals given an instantaneous function. The program was modified to solve the definite integral of the drainhole instantaneous source function. Tabulated results of dimensionless pressure vs time are given in Tables 1-4 for various dimensionless drainhole lengths and radii. These figures correspond to the pressure response at the center of a drainhole with dimensionless drainhole radius $r_{D_{dh}} = Y_D$. Type curves for the tabulated results are shown in Figs. 3-6.

Verification of Solution

Three cases found in the literature were considered for verification of the solution. These are³³⁻³⁵:

1. Partially Penetrating Uniform Fracture³³

$$\text{let } \frac{h_f}{x_f} \rightarrow 0$$

2. General Uniform Flux Fracture Solution³⁴

$$\text{let } L_D \rightarrow >10$$

3. Slanted Wellbore³⁵

$$\text{let } \theta_w \rightarrow 90^\circ$$

The partially-penetrating, uniform-flux fracture solution should approach the drainhole solution when the completion interval, h_f , relative to the fracture half length, x_f , approaches 0. Unfortunately, the values of $\frac{h_f}{x_f}$ presented by Raghavan were not small enough for the drainhole solution to approximate the partially-penetrating, uniform-flux fracture results.

The uniform flux, fully-penetrating vertical fracture solution was solved analytically by Gringarten and Ramey. Results discussed later show that the solution to a multiple vertical array of drainholes does approach the uniform flux vertical fracture solution as the number of drainholes in the reservoir increases.

Verification was obtained from the slanted wellbore solution although solutions were only presented for deviations from vertical up to 75° . Cinco defined the variable θ_w as the well deviation from vertical and plotted p_{wD} vs. t_D for various $h_D = \frac{h}{r_w}$. To match the drainhole solution, the slanted wellbore was approximated as a horizontal drainhole with the same horizontal projection as a well slanted at 75° . Fig. 7 shows the results of the match. The two solutions are within 1 % at all times. The results verify the drainhole solution and suggest that small variations from the horizontal do not significantly affect the pressure transient response.

SHORT-TIME APPROXIMATION

A short-time approximation may be derived by approximating an instantaneous source function for each of the three component systems as shown in Appendix B. The resulting equation is:

$$p_D = -\frac{1}{4L_D} Ei \left[-\frac{z_D^2 + y_D^2}{4t_D} \right] \quad (10)$$

This form of the short-time approximation is identical to the expression for radial flow when $x_D^2 + y_D^2$ is replaced by r_D^2 ,⁴⁴ indicating that early flow toward a drainhole is radial. The end of this radial flow period is given by Eq. (11).

$$t_{sD} \leq \min \left\{ \begin{array}{l} \frac{(1-x_D)^2}{20} \\ \left[Z_D + \frac{2}{L_D} (Z_{WD}-1) \right]^2 \\ \frac{20}{\left[Z_D + 2 \frac{Z_{WD}}{L_D} \right]^2} \end{array} \right\} \quad (11)$$

Combining the conditions for the natural logarithmic approximation with Eq. (11) defines the conditions that must be present for a graph of the short-time dimensionless pressure drop against the log of dimensionless time to be a straight line within 1%:

$$100(z_D^2 + y_D^2) < \min \left\{ \begin{array}{l} \frac{(1-x_D)^2}{20} \\ \frac{[z_D + 2 \frac{Z_{WD}}{L_D}]^2}{20} \\ \frac{[z_D + \frac{2}{L_D} (Z_{WD}-1)]^2}{20} \end{array} \right\} \quad (12)$$

If Eq.(12) is satisfied, a semilog straight line will begin at a t_D defined by the left side of Eq. (12), and end at a t_D defined by the right side. The short-time approximation has a slope of $0.5756L_D^{-1}$ when graphed on semilog graph paper. Thus, either the drainhole

half length or $\sqrt{\frac{k_z}{k_y}}$ can be determined if skin and storage effects become negligible before the end of the short time flow period. The short-time approximation is not a function of x_D , showing that this approximation is also the short-time solution for the infinite-conductivity drainhole.

If L_D is large, the initial radial flow ends quickly, and the short-time approximation in Eq. (10) is not accurate even at reasonably small times. Figure 3 shows that for such cases the early time log-log graph of P_D vs t_D is a straight line of slope $\frac{1}{2}$. Such behavior indicates that early flow is linear for large L_D .

LONG-TIME APPROXIMATION

The long-time behavior of the pressure transient response due to a uniform-flux drainhole was determined by referring to the instantaneous source functions for each of the three component systems in Fig. 2 used to generate the problem.

Derivation in Appendix C yields:

$$p_D(x_D, y_D, z_D, L_D, t_D) = \frac{\sqrt{\pi}}{4} \int_0^{t_D} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-n^2 \pi^2 L_D^2 t'_D \right] \cos n \pi z_{wD} \right. \\ \left. \cdot \cos n \pi (z_D L_D + z_{wD}) \right] \left[\operatorname{erf} \frac{(1+x_D)}{2\sqrt{t'_D}} + \operatorname{erf} \frac{(1-x_D)}{2\sqrt{t'_D}} \right] \frac{\exp \left[-\frac{y_D^2}{4t'_D} \right]}{\sqrt{t'_D}} dt'_D \\ + \int_{t_D}^{t_D} \frac{1}{2t_D} dt_D \quad (13)$$

Equation 13 applies when t_D is greater than that for the beginning of the pseudo-radial flow period, t_{ID} , as defined from Appendix C by:

$$t_{ID} = \max \left\{ \begin{array}{l} \frac{5}{\pi^2 L_D^2} \\ \frac{25 y_D^2}{3 \left(\frac{1}{2} + x_D \right)^2} \\ \frac{25}{3} \left(\frac{1}{2} - x_D \right)^2 \end{array} \right\} \quad (14)$$

Letting p_D equal the first term on the right, in Eq.(13), then the long-term approximation can be simplified to:

$$p_D = p_{ID} + \frac{1}{2} \ln \frac{t_D}{t_{ID}} \quad (15)$$

That late-time pressure drop is a constant plus $\frac{1}{2} \ln t_D$ indicates that flow is pseudo-radial after t_{ID} is reached. This is expected since all wellbore configurations in an infinite slab reservoir system should eventually reach a pseudo-radial flow period indicated by a 1.1515 semilog slope. Equation (14) can be used with type curve matching to determine if the appropriate semilog straight line was reached during a well test or to estimate well test duration requirements.

WELL TEST APPLICATIONS

The analysis of well test data for horizontal drainholes can be conducted with the type curves in Figs. 3-7 using conventional type curve matching procedures presented in the literature⁴⁶. Type curves for a particular drainhole radius can be generated using Eq. (9) and Simpson's Rule. The definitions of dimensionless drainhole variables suggest that the permeabilities in all three directions could be evaluated through the match point technique if all the other variables in the respective dimensionless group are known. The L_D match point yields the ratio of k_z to k_y and the p_D match point yields k_y . The x_D match point combined with other directional permeability data yields $\frac{k_y}{k_x}$. The $\phi \mu c$ constant may be determined by the time match, similarly to conventional type curve matching. Note that k_z can not be determined from well test data if $L_D > 10$ because flow in the vertical direction no longer influences pressure drop. This observation is reflected by the convergence of the pressure response lines to a single envelope of curves for $L_D > 10$.

In addition to the semilog techniques previously mentioned, well test data can also be used to determine the skin factor for the well.⁴⁶ The conditions for the start of the long-time flow approximations define the beginning of the appropriate semilog straight line. The magnitude of drainhole damage can be determined by subtracting the drain hole pseudo skin factor, to be presented from the total skin factor. The result yields a mechanical skin factor which is positive for a damaged drainhole, and negative for a stimulated drainhole.

EXTENSION TO MULTIPLE DRAINHOLES

Since multiple drainholes have also been drilled in some reservoirs^{1,25}, the pressure transient solution for a multiple drainhole system was investigated. A physical representation of the multiple drainhole system is given in Fig. 8. The system is identical to the single drainhole system except that several drainholes are present in a vertical array. In Fig. 8 the number of drainholes is three, although the solution is general for any number of drainholes.

The general solution for a single drainhole was obtained by multiplying together the instantaneous source functions for each of the three component systems:

$$S(x_D, y_D, z_D, t_D) = S(x_D, t_D) \cdot S(y_D, t_D) \cdot S(z_D, t_D) \quad (16)$$

The instantaneous source function in the z-direction was obtained by the method of images as described by Matthews et al⁴⁸. The configuration of the drainhole system in Fig. 1 is represented mathematically as an infinite array of drainholes stacked vertically a distance h apart as shown on the left side of Fig. 9. The solid lines between the drainholes are generated by the infinite array of drainholes, and represent no-flow boundaries in the reservoir. A no-flow boundary exists between each drainhole, and the choice of which two to represent the upper and lower reservoir boundaries defines the number of drainholes in the reservoir.

The solution for the multiple-drainhole case can be considered as m identical solutions to the single drainhole case stacked vertically where m is the number of drainholes. The multiple drainhole solution requires redefinition of only the dimensionless variables with respect to the number of drainholes in the reservoir. If L_D and z_{wD} are defined by Eqs.(17) and (18), then the dimensionless pressure for any point in the reservoir is given by Eq.(19).

$$L_D = \frac{mx_f}{h} \sqrt{\frac{k_z}{k_y}} \quad (17)$$

$$z_{wD} = \frac{z_{wm}}{h} \quad (18)$$

$$p_D(x_D, y_D, z_D, z_{wD}, L_D, t_D) = \frac{\sqrt{\pi}}{4} \int_0^{t_D} [1 + 2 \sum_{n=1}^{\infty} \exp(-n^2 \pi^2 L_D^2 t'_D)] \cos n \pi z_{wD} \cdot \cos n \pi (z_D L_D + z_{wD}) \left[\operatorname{erf} \frac{(1+x_D)}{2\sqrt{t'_D}} + \operatorname{erf} \frac{(1-x_D)}{2\sqrt{t'_D}} \right] \left[\frac{\exp\left(-\frac{y_D^2}{4t'_D}\right)}{\sqrt{t'_D}} \right] dt'_D \quad (19)$$

Equation (19) is the same equation as that for the single-drainhole solution, Eq.(9). Note that the definitions for dimensionless variables are also equivalent to the single-drainhole case when $m=1$.

Because the pressure response equation is the same as for a single drainhole, the results previously presented are also valid for the multiple-drainhole solution.

SIMILARITY TO VERTICAL FRACTURES

As the number of drainholes in a vertical array in a reservoir increases, the pressure transient solution to the drainhole problem should approach the solution for a uniform-flux vertical fracture case. It was noted previously that early flow into the drainhole is linear toward the drainhole if the dimensionless length of the drainhole is greater than 5 to 10. This behavior at short time is similar to a vertical fracture where flow is also linear in the y-direction. Intuitively, one would expect that later flow periods also represent the vertical fracture case.

The uniform flux vertical fracture solution is included with the type curve in Fig. 3. The two solutions are identical for $L_D > 10$. The similarity has also been proven mathematically⁵⁴.

$$p_D(x_D, y_D, z_D, z_{wD}, L_D, t_D) = \frac{\sqrt{\pi}}{4} \int_0^{t_D} \left[\operatorname{erf} \frac{(1+x_D)}{2\sqrt{t'_D}} + \operatorname{erf} \frac{(1-x_D)}{2\sqrt{t'_D}} \right] \left[\frac{\exp\left(-\frac{y_D^2}{4t'_D}\right)}{\sqrt{t'_D}} \right] dt'_D \quad (20)$$

when:

$$L_D > 10, \quad (21)$$

or;

$$\frac{m x_f}{h} \sqrt{\frac{k_z}{k_y}} > 10 \quad (22)$$

Eq.(20) is identical to the general expression for the uniform-flux vertical fracture as presented by Gringarten and Ramey³⁴. Eq. (22) shows that the uniform-flux vertical fracture pressure transient response can also be reproduced by increasing the number of drainholes.

Since the drainhole solution is identical to Gringarten and Ramey's solution for the vertical fracture, several special solutions can be directly inferred from that study. The expression for dimensionless pressure along the plane of a vertical fracture ($y_D=0$) is also the dimensionless pressure drop along the drainhole and is given by Eq.(23):

$$p_D(x_D, z_D, z_{wD}, L_D, t_D) = \frac{\sqrt{\pi}}{4} \int_0^{t_D} \frac{1}{\sqrt{t'_D}} \left[\operatorname{erf} \frac{(1+x_D)}{2\sqrt{t'_D}} + \operatorname{erf} \frac{(1-x_D)}{2\sqrt{t'_D}} \right] dt'_D \quad (23)$$

From the vertical fracture solution, the dimensionless pressure drop at $y_D=0$ and $z_D=0$ for a drainhole is given as:

$$p_D(x_D, t_D) = \frac{1}{2} (\pi t_D)^{\frac{1}{2}} \left[\operatorname{erf} \frac{(1+x_D)}{2\sqrt{t_D}} + \operatorname{erf} \frac{(1-x_D)}{2\sqrt{t_D}} \right] - \frac{(1-x_D)}{4} \operatorname{Ei} \left[-\frac{(1-x_D)^2}{4t_D} \right] - \frac{(1+x_D)}{4} \operatorname{Ei} \left[-\frac{(1+x_D)^2}{4t_D} \right] \quad (24)$$

The short-time approximation for large L_D can be defined directly from the vertical fracture solution as:

$$p_D(x_D, y_D, t_D) = \sqrt{\pi t_D} \exp\left(-\frac{y_D^2}{4t_D}\right) - |y_D| \operatorname{erfc}\left(\frac{|y_D|}{2\sqrt{t_D}}\right) \quad (25)$$

and along the drainhole ($y_D=0$):

$$p_D(x_D, t_D) = \sqrt{\pi t_D} \quad (26)$$

EXTENSION TO INFINITE-CONDUCTIVITY DRAINHOLES

The similarity between the drainhole and the vertical fracture cases suggests that other properties of the vertical fracture solution may be applicable to the drainhole solution. The pressure transient response to an infinite-conductivity drainhole should be considered since most drainholes are drilled and cased. Thus resistance to flow in the drainholes is negligible.

The infinite-conductivity solution for the fully-penetrating vertical fracture has been solved by several methods in the literature^{34,49,50}. It has been shown that the pressure drop along an infinite-conductivity vertical fracture can be obtained from the uniform-flux solution by calculating P_D for $y_D=0$ and $x_D=0.732$. The result is³⁴:

$$p_{wD}(t_D) = \frac{1}{2} \sqrt{\pi t_D} \left[\operatorname{erf}\left(\frac{0.134}{\sqrt{t_D}}\right) + \operatorname{erf}\left(\frac{0.866}{\sqrt{t_D}}\right) \right] - 0.067 \operatorname{Ei}\left(-\frac{0.018}{t_D}\right) - 0.433 \operatorname{Ei}\left(-\frac{0.759}{t_D}\right) \quad (27)$$

This equation was correct for all points along an infinite-conductivity vertical fracture for late and early times, and was assumed correct for all time. Of course the fracture result followed the previous result ($x_D=0.75$) obtained by Muskat⁵¹ for a well with partial penetration at steady state.

Equation (27) was derived by Gringarten and Ramey directly from the uniform-flux vertical fracture solution, Eq.(20). Consequently, Eq.(27) should also hold for infinite-conductivity drainhole with $L_D > 10.0$.

Fig. 5 shows a graph of the infinite-conductivity fracture solution is equivalent to the drainhole solution evaluated at $x_D=0.732$ when $L_D > 10$. By analogy, it is probable that the uniform-flux drainhole solution evaluated at $x_D=0.732$ also represents the dimensionless pressure along an infinite conductivity drainhole for L_D less than 10. Of course an infinite-conductivity drainhole could be represented a number of constant-flux segments and the flux along the drainhole determined as a function of time to prove this speculation. This step was beyond the objectives of this study.

DRAINHOLE PSEUDO SKIN FACTOR

A pseudo skin factor for drainholes was developed to compare drainhole productivity with that of vertical wells, and hydraulic fractures. Because the long-time flow for a drainhole has been shown to be pseudo-radial, it would be expected that a drainhole pseudo skin function should become constant at late time. The constant value reached by the pseudo skin function at late time has been defined⁵² as the pseudo skin factor, s , and is a function of space and system geometry, but not time. The pseudo skin factor for a uniform-flux drainhole system can be derived by analyzing the appropriate long-time approximations.

During late time the line source solution can be expressed as⁴⁴:

$$p_D = \frac{1}{2} \left[\ln \frac{t_D}{x_D^2 + y_D^2} + 0.80907 \right] \quad (28)$$

The long term approximation for the dimensionless pressure in Eq.(15) can be used with Eq.(28) to obtain the drainhole pseudo skin factor:

$$s = p_{wD} - \frac{1}{2} \ln t_D + \ln \frac{r_w}{x_f} - 0.4045 \quad (29)$$

Where p_{wD} and t_D are given by Eqs.(13) and (14).

Fig. 10 shows the pseudo skin factor at the wellbore graphed vs. L_D for various dimensionless drainhole radii, $\frac{r_{dh}}{x_f}$. The condition for negative pseudo skin factors is suggested by the following observation from Fig. 10:

$$\left[s + \ln \frac{x_f}{r_w} \right]_{at L_D=1/2} = \ln \frac{x_f}{r_{dh}} \quad (30)$$

Eq.(30) implies that if $r_w=r_{dh}$ then the pseudo skin factor for a drainhole is 0 when $L_D=\frac{1}{2}$. In general, the pseudo skin factor is negative if:

$$\frac{m}{h} x_f \sqrt{\frac{k_z}{k_y}} > \frac{1}{2} \quad (31)$$

In isotropic reservoirs, the pseudo skin factor is negative when the drainhole half-length is longer than the reservoir height. A large k_z should also help achieve the condition of Eq. 31. Such could be the case for reservoirs with natural fractures. Eq.(31) also suggests that reservoirs with thin beds could benefit from observations. Field results from France as reported by Giger et al.⁹ support these observations.

The three potentiometric studies²⁷⁻²⁹ of drainholes expressed results in terms of productivity increases. One of the principle observations from the studies was that increases in productivity are proportional to the combined length of all drainholes in the system until a point is reached where additional benefit per added foot begins to diminish²⁸. The negative unit slope in Fig. (10) confirms that productivity increases are directly proportional to the combined length of the drainholes. It also shows that the benefit to productivity does gradually decrease for each added foot beyond a certain amount. The potentiometric studies also observed that increases in drainhole length were more important than drainhole radius increases²⁹. Fig. 10 illustrates this observation and shows that productivity is not a function of drainhole radius for $L_D > 10$. A third observation from the potentiometric studies was that additional drainhole length was better than increasing the number of drainholes²⁷. Figure 10 shows that for $L_D > 10$, the only means for decreasing the pseudo skin factor is to increase the drainhole half length, L_D .

Figure 10 shows that for $L_D > 10$:

$$s = 1 + \ln \left(\frac{r_w}{x_f} \right) \quad (32)$$

This expression for the pseudo skin factor is equivalent to the pseudo skin factor for a uniform-flux vertical fracture³⁴. The apparent conclusion from Eq.(32) is that productivity increases from a drainhole may be as great as a uniform-flux vertical fracture of the same length for $L_D > 10$. The uniform-flux fracture case is very close to the infinite-fracture permeability case. Most hydraulic fractures have a finite conductivity that limits the effective fracture length⁵³. A drilled and cased drainhole should behave with infinite internal (wellbore) conductivity, although a skin effect may be present. Thus a drainhole may have a lower pseudo skin factor than a finite conductivity hydraulic fracture. Another observation is that if $k_y > k_x$, then fracture orientation is most likely in only the high permeability direction, while drainholes could be directionally drilled in the direction of low permeability for higher productivity. The orientation of a hydraulic fracture most likely occurs perpendicular to the principal axis of stress. If the reservoir contains naturally occurring fractures, hydraulic fracturing would be expected to form parallel to natural fractures and not enhance flow potential. Finally, it may be possible to obtain greater drainhole half lengths⁹ than with vertical fractures. Since one of the three previous observations will usually be pertinent, drainhole productivity may sometimes be greater than the productivity of hydraulic fractures in the same formation. Drainholes offer an alternative to hydraulic fracturing that has important and unique advantages in certain situations.

CONCLUSIONS

There are two possible types of transient pressure behavior depending on effective dimensionless drainhole half-length. If $L_D < 10$, flow is characterized by an initial radial flow perpendicular to the drainhole axis followed by a transition to a pseudo-radial flow

period. If $L_D > 10$, the initial radial flow period ends instantaneously for all practical purposes. Flow is then characterized by early time linear flow followed by a transition to late time pseudo-radial flow.

Log-log type curves are presented for various drainhole radii. Through type curve matches of p_D, t_D, L_D and x_D , permeability in three directions and $\phi\mu c$ can possibly be determined. The conditions for the start of the long time flow approximations define the beginning of the appropriate semilog straight line. Analysis of the semilog straight line will yield a skin factor which can be used with the drainhole pseudo skin factor to determine whether a drainhole is damaged or stimulated.

For cases where $L_D > 10$, the uniform flux drainhole solution matches the uniform flux vertical fracture solution. From the definition of L_D , the similarity between drainholes and vertical fractures is directly proportional to the ratios $\frac{x_f}{h}$ and $\sqrt{\frac{k_z}{k_y}}$.

The study demonstrated that the pressure transient response for multiple drainholes is identical to the single drainhole solution if dimensionless variables are defined relative to the number of drainholes. Consequently, a drainhole system will also approximate a vertical fracture if several drainholes are present.

Evaluation of the uniform flux drainhole solution for $x_D = 0.732$ showed that the drainhole pressure transient response for $L_D > 10$ was equivalent to the infinite conductivity vertical fracture. The early time solution for the infinite conductivity drainhole is given by the early time solution for the uniform flux case. The type curves presented can be used to determine reservoir characteristics and mechanical conditions similarly to the uniform flux drainhole case.

The drainhole pseudo skin factor defines situations where drainholes will be more productive than vertical wells. Drainholes are preferable to a vertical well if $\frac{x_f}{h} > \frac{1}{2}$. The performance of drainholes compared to vertical wells is further enhanced if $k_z > k_y$, or if the reservoir contains more than one drainhole.

The drainhole pseudo skin factor for $L_D > 10$ is identical to the pseudo skin factor for a uniform flux vertical fracture. In practice however, drainholes may have higher productivities than hydraulic fractures because (1) drainholes may be drilled to greater lengths than hydraulic fractures, (2) they can provide maximum productivity in reservoirs where $k_z \neq k_y$ through directional drainhole orientation, and (3) cased drainholes will have infinite wellbore conductivity acting, while a fracture of the same length will have some finite conductivity.

These statements are not intended to say that drainholes are always, or even sometimes, superior to a hydraulic fracture on the same well. Under certain conditions, the performance of a drainhole may exceed that of a hydraulic fracture on the same well. However, economics of the results has not been considered in this study. Horizontal directional drilling is difficult and costly. But the potential for improved well performance and improved well control make drain hole completions a very attractive target for the oil industry.

Obvious omissions in this study include consideration of a skin effect on the surface of the drainhole and wellbore (as well as drainhole) storage. Because of the similarity between drainhole and fracture solutions, it did not appear worthwhile to add the complexity of these issues. Neither have an important effect on the performance of high-conductivity fractures, and thus would have a small effect on the performance of a drainhole.

This study was actually inspired by field well test data that indicated a high-conductivity fracture was present in wells completed in an unconsolidated sand in Canada producing a heavy oil. One theory was that sand production of a complete sand-size distribution led to formation of "worm holes". We believed that a linear "worm hole" might appear very much like a high-conductivity vertical fracture, and set out on this study to answer an academic question. The answer is an unqualified conclusion that long "worm holes" (drainholes) can appear very much like vertical fractures as far as pressure transient data is concerned. This was not a major guess in the light of Cinco's work³⁵.

A surprising and perhaps more important finding, is that drainholes may exceed the performance of vertical fractures for some formations. For such formations, it is already known that there is a maximum effective fracture length for a specific fracture permeability-fracture width product. The added advantage of directional drilling the drainhole adds a new degree of freedom to such a well completion. This could have a major impact on anisotropic or fractured formations, and in EOR operations.

The question of comparative economics between fracturing and drainhole completions remains. However the obvious interest in drainholes in the recent literature signals that horizontal directional drilling and completions are here to stay. A major new tool is about to be added to drilling and production. The state of this art of engineering's time has come.

NOMENCLATURE

- c = Total reservoir compressibility
- h = Reservoir height
- h_D = Dimensionless reservoir height based on r_w
- h_f = Fracture height
- k_i = Permeability in the i direction
- L_D = Dimensionless drainhole half length
- M = Pressure point
- m = Number of drainholes in reservoir
- P = Source point
- p = Pressure
- p_D = Dimensionless pressure using k_y
- p_i = Initial reservoir pressure
- p_{wD} = Dimensionless Wellbore Pressure
- q = Flow rate per unit length of source
- q_w = Total flow rate
- r = radius
- S = Instantaneous source function
- s = Pseudo skin factor
- t = Time
- t_D = Dimensionless time using x_f and k_y
- t_{Dw} = Dimensionless time using r_w
- x, y, z = Space coordinates
- x_w, y_w, z_w = Drainhole center space coordinates
- x_f = Drainhole or Fracture half length
- z_e = Vertical distance between drainhole no-flow boundaries

GREEK SYMBOLS

- Δp = Pressure change from initial
- δ = Horizontal distance from the pressure point to the nearest drainhole tip
- η_i = Diffusivity constant in the i direction
- θ_w = Angle of well inclination from vertical
- μ = Viscosity
- ϕ = Porosity
- σ = Pseudo skin function
- τ = Integration variable

SUBSCRIPTS

- c = Critical
- D = Dimensionless
- dh = Drainhole
- f = Fracture
- l = Beginning of late time flow period
- s = End of short time flow period
- w = Well

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APPENDIX A

Solution Derivation

A general expression for pressure drop is:

$$\Delta p(M, t) = \frac{1}{\phi c} \int_0^t q(\tau) S(M, t - \tau) d\tau \quad (A-1)$$

In Eq. (A-1) S is the instantaneous uniform flux source function for the particular source-reservoir system. Referring to Fig. 2, the first source is a slab source infinite in the y and z directions and of thickness $2x_f$. The instantaneous source function for an infinite slab source is given in Ref. 36 as:

$$S(x, t) = \frac{1}{2} \left[\operatorname{erf} \frac{\frac{x_f}{2} + (x - x_w)}{2\sqrt{\eta_x t}} + \operatorname{erf} \frac{\frac{x_f}{2} - (x - x_w)}{2\sqrt{\eta_x t}} \right] \quad (A-2)$$

The second instantaneous source is that for an infinite plane source in the x - z plane. The instantaneous source function for an infinite plane source is given by Ref. 36 as:

$$S(y, t) = \frac{\exp \left[-\frac{(y - y_w)^2}{4\eta_y t} \right]}{2\sqrt{\pi\eta_y t}} \quad (A-3)$$

The third instantaneous source function is that for an infinite plane source in a slab reservoir. This is also presented by Gringarten and Ramey³⁶ as:

$$S(z, t) = \frac{1}{h} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-\frac{n^2 \pi^2 \eta_z t}{h^2} \right] \cos n\pi \frac{z_w}{h} \cos n\pi \frac{z}{h} \right] \quad (A-4)$$

Fig. 2 shows that the intersection of Eqs. (A-2), (A-3), and (A-4) is the necessary configuration describing the drainhole system. By

the principle of the Newman product method, the instantaneous source function for the drainhole system is given as the product of the three individual source functions. The instantaneous source function for the drainhole system is:

$$S(x,y,z,t) = \frac{1}{h} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-\frac{n^2 \pi^2 \eta_z t}{h^2} \right] \cos n \pi \frac{z_w}{h} \cos n \pi \frac{z}{h} \right] \cdot \frac{1}{2} \left[\operatorname{erf} \frac{\frac{x_f}{2} + (x-x_w)}{2\sqrt{\eta_x t}} + \operatorname{erf} \frac{\frac{x_f}{2} - (x-x_w)}{2\sqrt{\eta_x t}} \right] \frac{\exp \left[-\frac{(y-y_w)^2}{4\eta_y t} \right]}{2\sqrt{\pi \eta_y t}} \quad (\text{A-5})$$

Substituting Eq.(A-5) into Eq.(A-1) yields the pressure drop at any point M due to the drainhole:

$$\Delta p(x,y,z,t) = \frac{1}{\phi c} \int_0^t q(\tau) \frac{1}{h} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-\frac{n^2 \pi^2 \eta_z \tau}{h^2} \right] \cos n \pi \frac{z_w}{h} \cos n \pi \frac{z}{h} \right] \cdot \frac{1}{2} \left[\operatorname{erf} \frac{\frac{x_f}{2} + (x-x_w)}{2\sqrt{\eta_x \tau}} + \operatorname{erf} \frac{\frac{x_f}{2} - (x-x_w)}{2\sqrt{\eta_x \tau}} \right] \frac{\exp \left[-\frac{(y-y_w)^2}{4\eta_y \tau} \right]}{2\sqrt{\pi \eta_y \tau}} d\tau \quad (\text{A-6})$$

where

$$q_w = 2x_f q \quad (\text{A-7})$$

The total flowrate, q_w , is assumed constant, and the flowrate per unit length, $q(\tau)$, can be removed from the integral.

The relation between Δp and p_D is given by Eq. (A-8).

$$p_D = \frac{2\pi k_y h \Delta p}{q_w \mu} \quad (\text{A-8})$$

Combining Eqs. (A-1), (A-8) and the dimensionless variable definitions yields the dimensionless pressure drop at any point in the drainhole system given by Eq. (9).

APPENDIX B

Short-Time Approximation Derivation

For small values of t_D the expression:

$$\operatorname{erf} \left[\frac{x}{2\sqrt{t_D}} \right] \rightarrow \pm 1 \quad (\text{B-1})$$

approaches unity for positive values of x , and approaches negative one for negative values of x . Thus, the instantaneous source function given by Eq. (A-2) is unity at early time. The proper time limit for this approximation is presented by Gringarten and Ramey³⁰ as:

$$t \leq \frac{\delta^2}{20\eta_x} \quad \text{for } |x| < x_f \quad (\text{B-2})$$

where δ is defined as the minimum distance to the nearest source boundary. For the drainhole problem, the minimum distance is the

distance from x to x_f .

$$\delta = |x - x_f| \quad (\text{B-3})$$

From Eq.(B-2) the short time approximation for the dimensionless instantaneous source function is given by Eqs.(B-4),(B-5), and (B-6) for points between the tips of the drainhole, on the tips, and beyond the drainhole, respectively.

$$S(x,t) = 1 \quad (\text{B-4})$$

$$S(x,t) = \frac{1}{2} \quad (\text{B-5})$$

$$S(x,t) = 0 \quad (\text{B-6})$$

At early times, there is no flow in the reservoir beyond the edges of the drainhole from Eq.(B-6). The appropriate time limit maybe obtained by combining Eqs.(B-2) and (B-3). This maximum time is the end of the short-time flow period, t_{sD} , and is given in Eqs.(B-7) and (B-8).

$$t_{sD} \leq \frac{(1-x_D)^2}{20} \quad x_D \neq 1 \quad (\text{B-7})$$

$$t_{sD} \leq \frac{1}{20} \quad x_D = 1 \quad (\text{B-8})$$

The instantaneous source function for an infinite plane in a slab reservoir, Eq.(A-4), can be approximated at short time by Eq.(B-9).^{36,43}

$$S(z,t) = \frac{1}{2\sqrt{\pi \eta_z t}} \exp \left[-\frac{(z-z_w)^2}{4\eta_z t} \right] \quad (\text{B-9})$$

when:

$$\max \left\{ \frac{\eta_z t}{(z+z_w)^2}, \frac{\eta_z t}{(z+z_w-2h)^2} \right\} \leq \frac{1}{20} \quad (\text{B-10})$$

Substituting the definitions for dimensionless variables from Eqs.(3) through (8) into Eq.(B.9) yields the following dimensionless short-time approximation of Eq.(A-4):

$$S(Z_D, t_D) = \sqrt{\frac{k_y}{k_z}} \cdot \frac{1}{2x_f \sqrt{\pi t_D}} \exp \left[-\frac{z_D^2}{4t_D} \right] \quad (\text{B-11})$$

when:

$$t_{zD} \leq \min \left\{ \frac{[z_D + \frac{2}{L_D}(z_w D - 1)]^2}{20}, \frac{[z_D + 2\frac{z_w D}{L_D}]^2}{20} \right\} \quad (\text{B-12})$$

The short-time approximation for the drainhole system is the product of the three short-time instantaneous source approximations, Eqs.(B-11), (B-4) and (A-4). The result is Eq.(B-13):

$$S(x_D, y_D, z_D, t_D) = \sqrt{\frac{k_y}{k_z}} \frac{1}{2x_f \sqrt{\pi t_D}} \exp\left[-\frac{z_D^2}{4t_D}\right] \cdot \frac{1}{2x_f \sqrt{\pi t_D}} \exp\left[-\frac{y_D^2}{4t_D}\right] \quad (\text{B-13})$$

Substituting Eq.(B-13) into Eq.(9) yields:

$$p_D(y_D, z_D, t_D) = \frac{h}{4x_f} \sqrt{\frac{k_y}{k_z}} \int_0^{t_D} \frac{\exp\left[-\frac{z_D^2 + y_D^2}{4t'_D}\right]}{t'_D} dt'_D \quad (\text{B-14})$$

Eq.(B-14) can be expressed in terms of the exponential integral as shown by Eq.(10).

APPENDIX C

Long-Time Approximation Derivation:

The instantaneous source function for an infinite plane source in an infinite slab reservoir, Eq.(A-4), can be approximated³⁶ within 1 % by $\frac{1}{h}$ in Eq.(C-1):

$$S(z, t) = \frac{1}{h} \left[1 + 2 \sum_{n=1}^{\infty} \exp\left[-\frac{n^2 \pi^2 \eta_z t}{h^2}\right] \cos n\pi \frac{z_w}{h} \cos n\pi \frac{z}{h} \right] \rightarrow \frac{1}{h} \quad (\text{C-1})$$

when:

$$t \geq \frac{5h^2}{\pi^2 \eta_z}, \quad (\text{C-2})$$

or:

$$t_D \geq \frac{5}{\pi^2 L_D^2} \quad (\text{C-2a})$$

The instantaneous source function for an infinite plane source, Eq.(A-3), can be approximated by expanding the exponential term⁴⁵ as in Eq.(C-3):

$$S(y, t) = \frac{\exp\left[-\frac{(y-y_w)^2}{4\eta_y t}\right]}{2\sqrt{\pi \eta_y t}} = \frac{1}{2x_f \sqrt{\pi t_D}} \left[1 - \frac{y_D^2}{4t_D} + \frac{y_D^2}{32t_D^2} + O\left(\frac{1}{t_D^3}\right) \right] \quad (\text{C-3})$$

The left hand side of Eq.(3.3) can be approximated by the first term in the expansion when:

$$\frac{1}{2x_f \sqrt{\pi t_D}} \geq \frac{100y_D^2}{4t_D \cdot 2x_f \sqrt{\pi t_D}} \quad (\text{C-4})$$

or:

$$t_D \geq 25y_D^2 \quad (\text{C-5})$$

The third instantaneous source function, Eq.(A-2), can be approximated by expansion of the error function⁴⁵.

$$\operatorname{erf}\left[\frac{\frac{1}{2} + x_D}{2\sqrt{t_D}}\right] = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(2n+1)} \left[\frac{\frac{1}{2} + x_D}{2\sqrt{t_D}}\right]^{2n+1}$$

$$= \frac{\frac{1}{2} + x_D}{\sqrt{\pi t_D}} - \frac{(\frac{1}{2} + x_D)^3}{12\sqrt{\pi t_D^3}} + O\left[\frac{1}{t_D^{5/2}}\right] \quad (\text{C-6})$$

The first term is an accurate expression of the error function within 1% when:

$$\frac{\frac{1}{2} + x_D}{\sqrt{\pi t_D}} \geq \frac{100(\frac{1}{2} + x_D)^3}{12\sqrt{\pi t_D^3}} \quad (\text{C-7})$$

or:

$$t_D \geq \frac{25}{3} \left(\frac{1}{2} + x_D\right)^2 \quad (\text{C-8})$$

Similarly, the second error function is approximated within one percent by:

$$\operatorname{erf}\left[\frac{\frac{1}{2} - x_D}{2\sqrt{t_D}}\right] = \frac{\frac{1}{2} - x_D}{\sqrt{\pi t_D}} \quad (\text{C-9})$$

when:

$$t_D \geq \frac{25}{3} \left(\frac{1}{2} - x_D\right)^2 \quad (\text{C-10})$$

The instantaneous source function for a drainhole may be determined by substituting Eqs.(C-1), (C-3), (C-6), and (C-9) into the general expression for the drainhole instantaneous source function, Eq.(A-5):

$$S(X_D, Y_D, Z_D, Z_{wD}, L_D, t_D) = \frac{1}{h} \cdot \frac{1}{2x_f \sqrt{\pi t_D}} \cdot \frac{1}{2} \left[\frac{\frac{1}{2} + x_D}{\sqrt{\pi t_D}} + \frac{\frac{1}{2} - x_D}{\sqrt{\pi t_D}} \right] \quad (\text{C-11})$$

The dimensionless pressure long-time approximation is then found by combining Eq.(C-11) with Eq.(9) to obtain Eq. (13).

APPENDIX D

Multiple Drainhole Solution Derivation

The dimensionless pressure drop can be defined in terms of the total height of the reservoir, h , and the total flowrate from all the drainholes, q_w .

$$p_D = \frac{2\pi k_y h \Delta p}{q_w \mu} \quad (\text{D-1})$$

where

$$h = z_e \text{ m} \quad (\text{D-2})$$

$$q_w = 2 x_f m q \quad (D-3)$$

Substituting Eqs.(D.2) and (D.3) into Eq.(D.1) shows that the m 's cancel and the dimensionless pressure, as defined, is equal to the definition for dimensionless pressure in Eq.(A-8).

The pressure transient response is given by replacing h with z_e in Eq.(9):

$$\Delta p(x,y,z,t) = \frac{1}{\phi c} \int_0^t q(\tau) \frac{1}{z_e} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-\frac{n^2 \pi^2 \eta_z \tau}{z_e^2} \right] \cos n\pi \frac{z_w}{z_e} \cos n\pi \frac{z}{z_e} \right] \\ \cdot \frac{1}{2} \left[\operatorname{erf} \frac{\frac{x_f}{2} + (x-x_w)}{2\sqrt{\eta_x \tau}} + \operatorname{erf} \frac{\frac{x_f}{2} - (x-x_w)}{2\sqrt{\eta_x \tau}} \right] \frac{\exp \left[-\frac{(y-y_w)^2}{4\eta_y \tau} \right]}{2\sqrt{\pi \eta_y \tau}} d\tau \quad (D-4)$$

Substituting Eqs.(D.1-3) into Eq.(D.4) yields:

$$p_D = \frac{2\pi k_y h m}{m \mu \phi c x_f h} \int_0^t \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-\frac{n^2 \pi^2 \eta_z \tau m^2}{h^2} \right] \cos n\pi \frac{z_w m}{h} \cos n\pi \frac{z m}{h} \right] \\ \left[\operatorname{erf} \frac{\frac{x_f}{2} + (x-x_w)}{2\sqrt{\eta_x \tau}} + \operatorname{erf} \frac{\frac{x_f}{2} - (x-x_w)}{2\sqrt{\eta_x \tau}} \right] \frac{\exp \left[-\frac{(y-y_w)^2}{4\eta_y \tau} \right]}{2\sqrt{\pi \eta_y \tau}} d\tau \quad (D-5)$$

Eq.(D.5) suggests that the dimensionless variables x_D , y_D , z_D , and t_D should still be as previously defined. Thus:

$$p_D = \frac{\sqrt{\pi}}{4} \int_0^{t_D} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left[-\frac{n^2 \pi^2 \eta_z' t_D x_f^2 m^2}{\eta_y h^2} \right] \cos n\pi \frac{z_w m}{h} \cos n\pi \frac{z m}{h} \right] \\ \cdot \left[\operatorname{erf} \frac{1+x_D}{2\sqrt{t_D}} + \operatorname{erf} \frac{1-x_D}{2\sqrt{t_D}} \right] \cdot \left[\frac{\exp \left[-\frac{y_D^2}{4t_D'} \right]}{\sqrt{t_D'}} \right] dt_D' \quad (D-6)$$

If L_D and Z_{wD} are defined by Eqs. (17) and (18) then Eq. (19) follows.

TABLE 1 - Dimensionless Pressure for a Uniform Flux Drainhole in an Infinite Slab Reservoir
for $x_D=0$ and $r_{Ddh}=0.0005$

	XD = 0.0 YD = 0.5e-03 ZD = 0.0 ZWH = 0.5000						
LD TD	1.250d-01	2.500d-01	5.000d-01	1.000d+00	2.500d+00	1.000d+01	5.000d+01
1.00e-04	1.3529e+01	6.8026e+00	3.4014e+00	1.7007e+00	6.8027e-01	1.7007e-01	3.6243e-02
1.00e-03	1.8135e+01	9.1054e+00	4.5528e+00	2.2764e+00	9.1056e-01	2.2888e-01	7.4564e-02
1.00e-02	2.2741e+01	1.1409e+01	5.7045e+00	2.8522e+00	1.1416e+00	3.5034e-01	1.9577e-01
1.00e-01	2.7334e+01	1.3705e+01	6.8527e+00	3.4386e+00	1.5285e+00	7.3188e-01	5.7731e-01
1.00e+00	3.0913e+01	1.5499e+01	7.8863e+00	4.3271e+00	2.4146e+00	1.6180e+00	1.4634e+00
1.00e+01	3.2472e+01	1.6639e+01	9.0019e+00	5.4427e+00	3.5302e+00	2.7336e+00	2.5790e+00
1.00e+02	3.3620e+01	1.7787e+01	1.0150e+01	6.5907e+00	4.6782e+00	3.8816e+00	3.7270e+00
1.00e+03	3.4772e+01	1.8938e+01	1.1301e+01	7.7420e+00	5.8295e+00	5.0329e+00	4.8783e+00
1.00e+04	3.5923e+01	2.0090e+01	1.2453e+01	8.8937e+00	6.9812e+00	6.1845e+00	6.0300e+00
1.00e+05	3.7075e+01	2.1242e+01	1.3605e+01	1.0045e+01	8.1329e+00	7.3363e+00	7.1817e+00
1.00e+06	3.8227e+01	2.2393e+01	1.4756e+01	1.1197e+01	9.2846e+00	8.4880e+00	8.3334e+00
1.00e+07	3.9378e+01	2.3545e+01	1.5908e+01	1.2349e+01	1.0436e+01	9.6397e+00	9.4851e+00

TABLE 2 - Dimensionless Pressure for a Uniform Flux Drainhole in an Infinite Slab Reservoir
for $x_D=0$ and $r_{Ddh}=0.005$

	XD = 0.0 YD = 0.5e-02 ZD = 0.0 ZWH = 0.5000						
LD TD	1.250d-01	2.500d-01	5.000d-01	1.000d+00	2.500d+00	1.000d+01	5.000d+01
1.00e-04	4.5126e+00	2.2571e+00	1.1285e+00	5.6426e-01	2.2571e-01	5.6426e-02	1.3298e-02
1.00e-03	9.0086e+00	4.5051e+00	2.2525e+00	1.1263e+00	4.5051e-01	1.1386e-01	5.0880e-02
1.00e-02	1.3604e+01	6.8029e+00	3.4014e+00	1.7007e+00	6.8104e-01	2.3508e-01	1.7185e-01
1.00e-01	1.8196e+01	9.0987e+00	4.5494e+00	2.2870e+00	1.0678e+00	6.1654e-01	5.5331e-01
1.00e+00	2.1774e+01	1.0892e+01	5.5830e+00	3.1754e+00	1.9539e+00	1.5026e+00	1.4394e+00
1.00e+01	2.3334e+01	1.2032e+01	6.6986e+00	4.2911e+00	3.0695e+00	2.6182e+00	2.5550e+00
1.00e+02	2.4482e+01	1.3180e+01	7.8465e+00	5.4390e+00	4.2175e+00	3.7662e+00	3.7030e+00
1.00e+03	2.5633e+01	1.4332e+01	8.9979e+00	6.5904e+00	5.3688e+00	4.9175e+00	4.8543e+00
1.00e+04	2.6785e+01	1.5483e+01	1.0150e+01	7.7420e+00	6.5205e+00	6.0692e+00	6.0060e+00
1.00e+05	2.7936e+01	1.6635e+01	1.1301e+01	8.8937e+00	7.6722e+00	7.2209e+00	7.1577e+00
1.00e+06	2.9088e+01	1.7787e+01	1.2453e+01	1.0045e+01	8.8239e+00	8.3726e+00	8.3094e+00
1.00e+07	3.0240e+01	1.8938e+01	1.3605e+01	1.1197e+01	9.9756e+00	9.5243e+00	9.4611e+00

TABLE 3 - Dimensionless Pressure for a Uniform Flux Drainhole in an Infinite Slab Reservoir
for $x_D=0.732$ and $r_{Ddh}=0.0005$

XD = 0.732 YD = 0.5e-03 ZD = 0.0 ZWH = 0.5000

LD TD	1.250d-01	2.500d-01	5.000d-01	1.000d+00	2.500d+00	1.000d+01	5.000d+01
1.00e-04	1.3529e+01	6.8026e+00	3.4014e+00	1.7007e+00	6.8027e-01	1.7007e-01	3.6243e-02
1.00e-03	1.8135e+01	9.1054e+00	4.5528e+00	2.2764e+00	9.1056e-01	2.2888e-01	7.4564e-02
1.00e-02	2.2721e+01	1.1399e+01	5.6994e+00	2.8497e+00	1.1406e+00	3.4956e-01	1.9498e-01
1.00e-01	2.6657e+01	1.3366e+01	6.6833e+00	3.3510e+00	1.4638e+00	6.6767e-01	5.1310e-01
1.00e+00	2.9461e+01	1.4773e+01	7.5022e+00	4.0614e+00	2.1724e+00	1.3763e+00	1.2218e+00
1.00e+01	3.0938e+01	1.5860e+01	8.5667e+00	5.1260e+00	3.2370e+00	2.4409e+00	2.2863e+00
1.00e+02	3.2080e+01	1.7002e+01	9.7087e+00	6.2680e+00	4.3790e+00	3.5829e+00	3.4283e+00
1.00e+03	3.3231e+01	1.8152e+01	1.0859e+01	7.4187e+00	5.5297e+00	4.7336e+00	4.5791e+00
1.00e+04	3.4383e+01	1.9304e+01	1.2011e+01	8.5704e+00	6.6813e+00	5.8852e+00	5.7307e+00
1.00e+05	3.5534e+01	2.0456e+01	1.3163e+01	9.7220e+00	7.8330e+00	7.0369e+00	6.8824e+00
1.00e+06	3.6686e+01	2.1607e+01	1.4314e+01	1.0874e+01	8.9847e+00	8.1886e+00	8.0341e+00
1.00e+07	3.7838e+01	2.2759e+01	1.5466e+01	1.2025e+01	1.0136e+01	9.3403e+00	9.1858e+00

TABLE 4 - Dimensionless Pressure for a Uniform Flux Drainhole in an Infinite Slab Reservoir
for $x_D=0.732$ and $r_{Ddh}=0.005$

XD = 0.732 YD = 0.5e-02 ZD = 0.0 ZWH = 0.5000

LD TD	1.250d-01	2.500d-01	5.000d-01	1.000d+00	2.500d+00	1.000d+01	5.000d+01
1.00e-04	4.5126e+00	2.2571e+00	1.1285e+00	5.6426e-01	2.2571e-01	5.6426e-02	1.3298e-02
1.00e-03	9.0086e+00	4.5051e+00	2.2525e+00	1.1263e+00	4.5051e-01	1.1386e-01	5.0880e-02
1.00e-02	1.3584e+01	6.7927e+00	3.3964e+00	1.6982e+00	6.8001e-01	2.3430e-01	1.7106e-01
1.00e-01	1.7518e+01	8.7600e+00	4.3800e+00	2.1994e+00	1.0031e+00	5.5234e-01	4.8911e-01
1.00e+00	2.0323e+01	1.0166e+01	5.1989e+00	2.9098e+00	1.7117e+00	1.2610e+00	1.1978e+00
1.00e+01	2.1800e+01	1.1253e+01	6.2634e+00	3.9743e+00	2.7763e+00	2.3255e+00	2.2623e+00
1.00e+02	2.2942e+01	1.2395e+01	7.4054e+00	5.1164e+00	3.9183e+00	3.4676e+00	3.4043e+00
1.00e+03	2.4093e+01	1.3546e+01	8.5562e+00	6.2671e+00	5.0690e+00	4.6183e+00	4.5550e+00
1.00e+04	2.5245e+01	1.4697e+01	9.7078e+00	7.4187e+00	6.2206e+00	5.7699e+00	5.7067e+00
1.00e+05	2.6396e+01	1.5849e+01	1.0859e+01	8.5704e+00	7.3723e+00	6.9216e+00	6.8583e+00
1.00e+06	2.7548e+01	1.7001e+01	1.2011e+01	9.7221e+00	8.5240e+00	8.0733e+00	8.0101e+00
1.00e+07	2.8700e+01	1.8153e+01	1.3163e+01	1.0874e+01	9.6757e+00	9.2250e+00	9.1618e+00

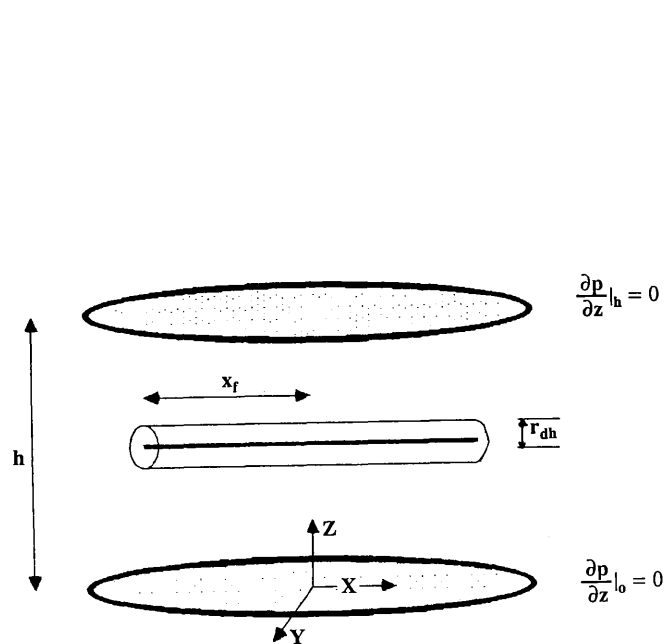


Fig. 1—Physical representation of single drainhole system.

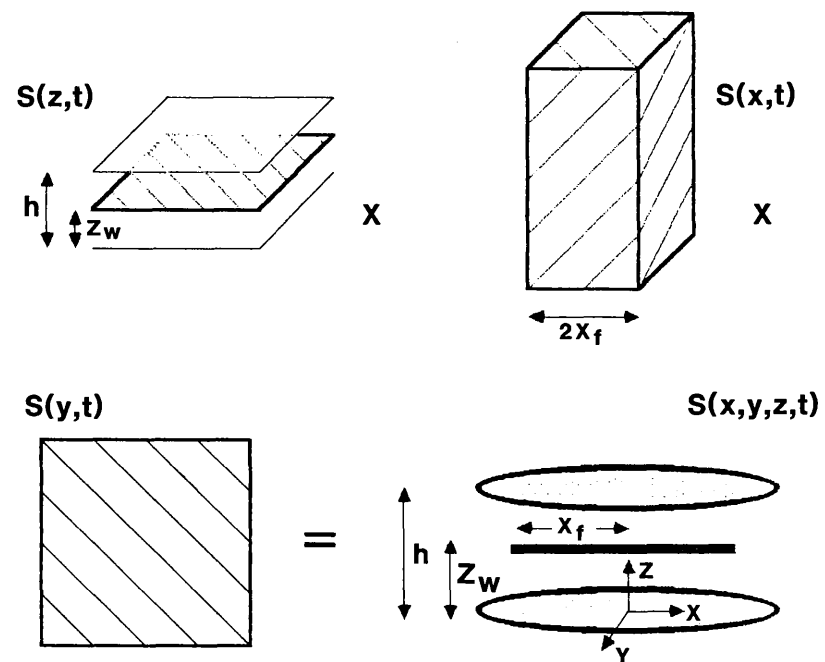


Fig. 2—Newman product representation of single drainhole system.

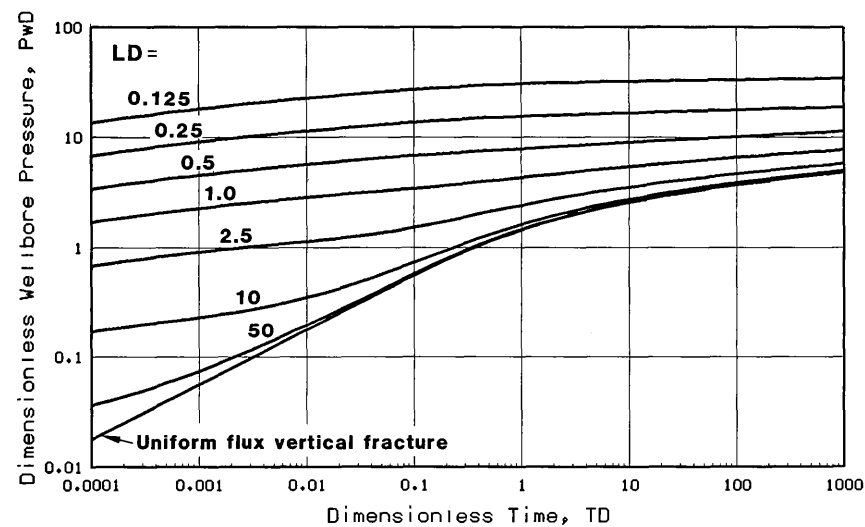


Fig. 3— p_{wD} vs. t_D for a uniform flux drainhole in an infinite slab reservoir for $x_D = 0$ and $r_{Ddh} = 0.0005$.

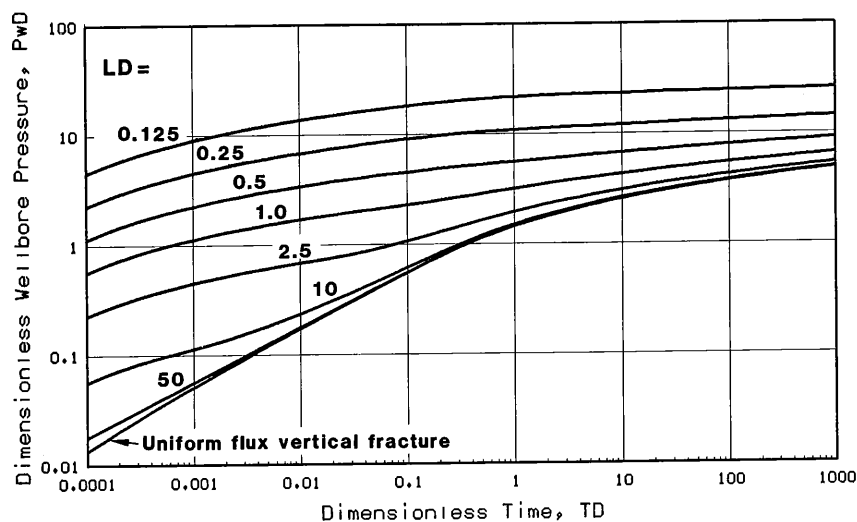


Fig. 4— p_{wD} vs. t_D for a uniform flux drainhole in an infinite slab reservoir for $x_D = 0$ and $r_{Ddh} = 0.005$.

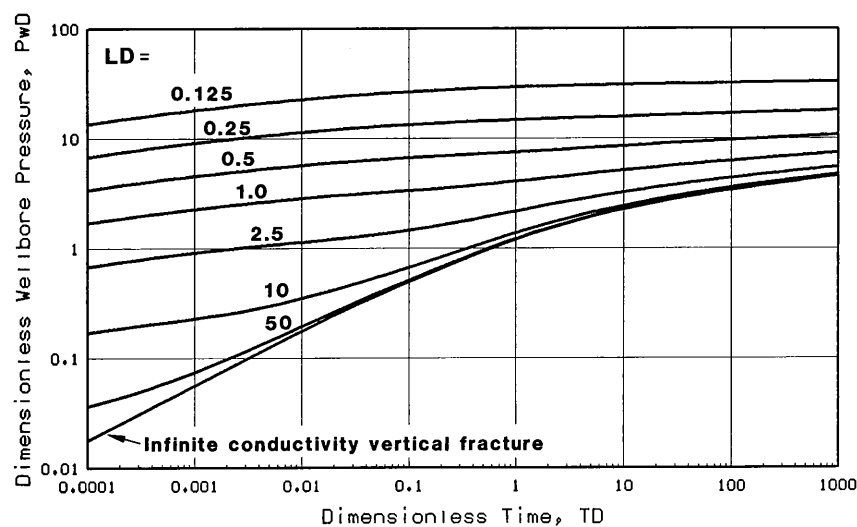


Fig. 5— p_{wD} vs. t_D for a uniform flux drainhole in an infinite slab reservoir for $x_D = 0.732$ and $r_{Ddh} = 0.0005$.

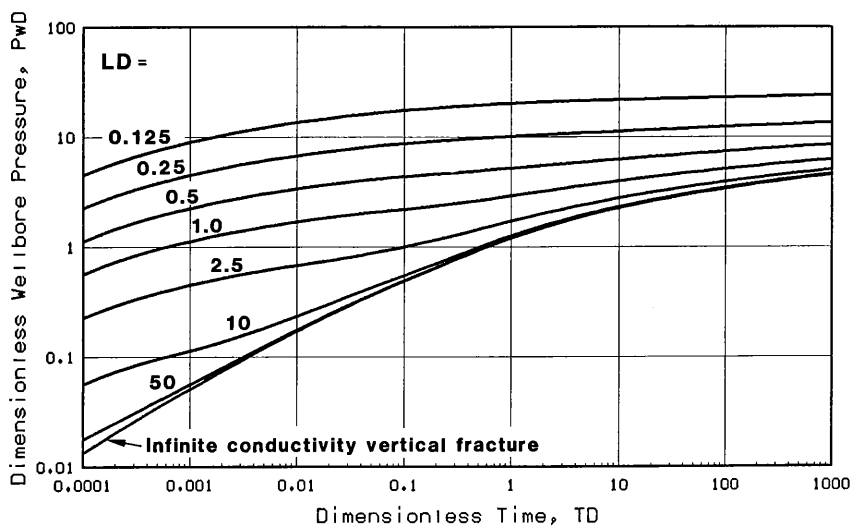


Fig. 6— p_{wD} vs. t_D for a uniform flux drainhole in an infinite slab reservoir for $x_D = 0.732$ and $r_{Ddh} = 0.005$.

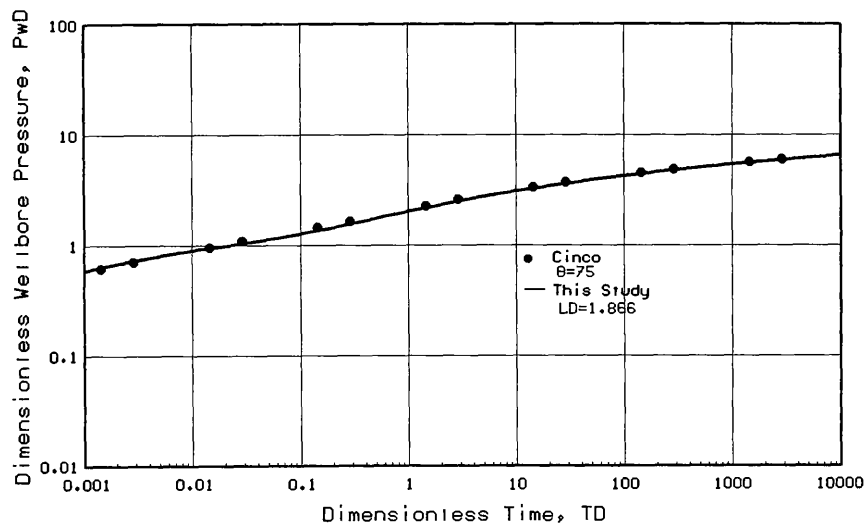


Fig. 7—Comparison of drainhole solution and Cinco solution for a 75° slanted wellbore.

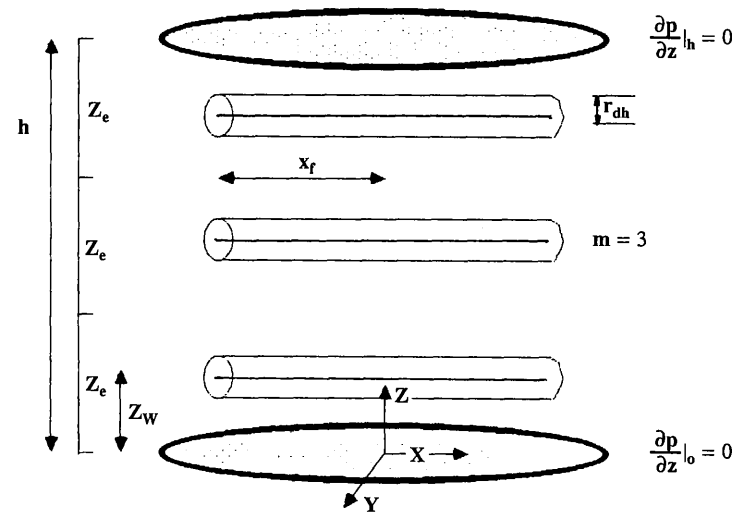


Fig. 8—Physical representation of multiple drainhole system.

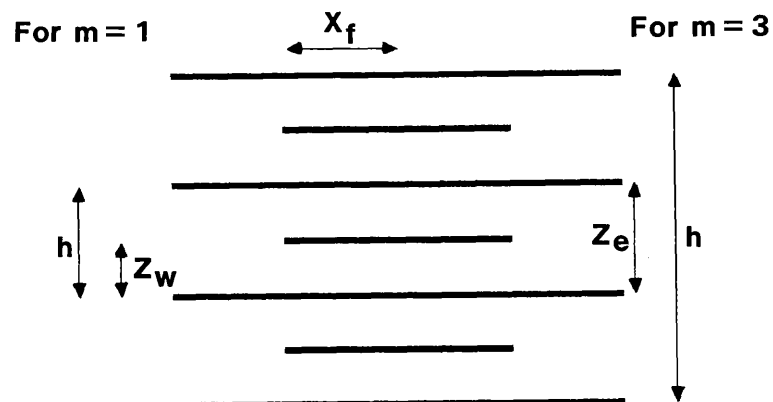


Fig. 9—Newman product representation of multiple drainhole system.

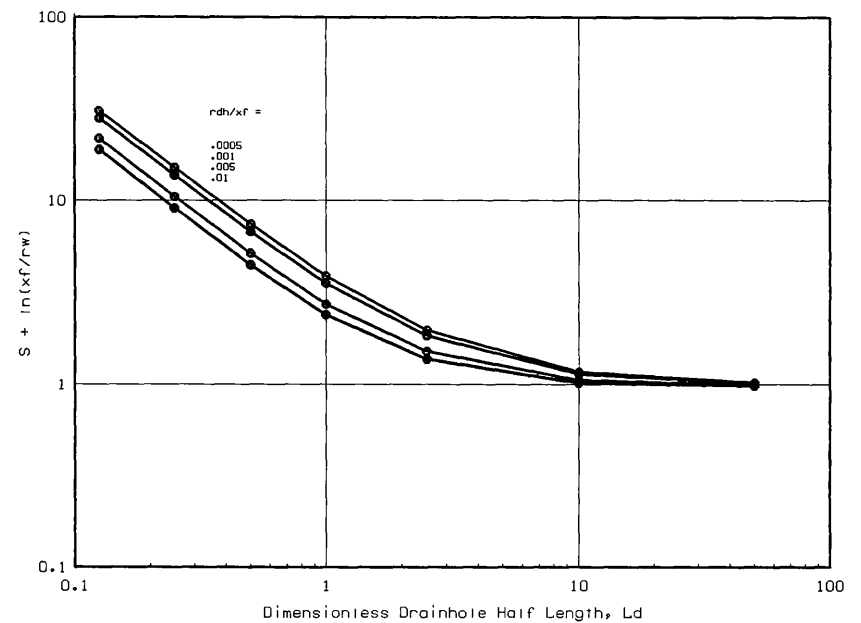


Fig. 10—Uniform flux drainhole pseudo skin factor vs. dimensionless drainhole half length for various r_{Ddh} .